

Ex.no:2	EDA-DATA IMPORT AND EXPORT

Aim:

To inspect the Life Expectancy dataset features, detect null values, and export data into CSV, Excel, and SQLite database formats.

Algorithm:

1. Import pandas and sqlite3 libraries.
2. Read Life Expectancy Data.csv using `pd.read_csv()`.
3. Inspect data structure using `.head()`, `.tail()`, `.info()`, `.describe()`, and `.isnull().sum()`.
4. Export DataFrame to .csv and .xlsx formats using `.to_csv()` and `.to_excel()`.
5. Connect to `life_expectancy.db` using `sqlite3` and write data into an SQL table via `.to_sql()`.
6. Close the database connection and end execution.

Program:

```
import os
import pandas as pd
import sqlite3

# Dataset path
path = "/content"

print("Dataset path:", path)

print("\nFiles in dataset:")
print(os.listdir(path))

# Find CSV file automatically
csv_files = [f for f in os.listdir(path) if f.lower().endswith(".csv")]

if not csv_files:
    raise FileNotFoundError("No CSV file found in the dataset path.")

csv_file = csv_files[0]

print("\nCSV file:", csv_file)

# Load dataset
df = pd.read_csv(
    os.path.join(path, csv_file),
    encoding="latin1",
    low_memory=False
)

print("\nFirst 5 rows:")
display(df.head())

print("\nLast 5 rows:")
display(df.tail())
```

```
print("\nDataset information:")
df.info()

print("\nShape:")
print(df.shape)

print("\nColumns:")
print(df.columns.tolist())

print("\nData types:")
print(df.dtypes)

print("\nStatistical description:")
display(df.describe(include="all").T)

print("\nMissing values:")
display(df.isnull().sum().sort_values(ascending=False))

print("\nDuplicate rows:")
print(df.duplicated().sum())

print("\nStatus:")
display(df["Status"].value_counts())

print("\nCountries:")
print(df["Country"].nunique())

print("\nYears:")
display(df["Year"].value_counts())

print("\nTop 10 countries by life expectancy:")
display(
    df.groupby("Country")["Life expectancy"]
      .mean()
      .sort_values(ascending=False)
      .head(10)
)

print("\nBottom 10 countries by life expectancy:")
display(
    df.groupby("Country")["Life expectancy"]
      .mean()
      .sort_values()
      .head(10)
)

print("\nCorrelation:")
display(df.corr(numeric_only=True))

# Excel
df.to_excel("life_expectancy_data.xlsx", index=False)

# JSON
df.to_json("life_expectancy_data.json", orient="records", indent=4)

# CSV
```

```
df.to_csv("life_expectancy_data.csv", index=False)

print("\nCSV, Excel and JSON files created successfully.")
```

Output:

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2938 entries, 0 to 2937
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Country                               2938 non-null   object
1   Year                                  2938 non-null   int64
2   Status                                2938 non-null   object
3   Life expectancy                       2928 non-null   float64
4   Adult Mortality                       2928 non-null   float64
5   infant deaths                         2938 non-null   int64
6   Alcohol                               2744 non-null   float64
7   percentage expenditure                 2938 non-null   float64
8   Hepatitis B                           2385 non-null   float64
9   Measles                               2938 non-null   int64
10  BMI                                    2904 non-null   float64
11  under-five deaths                     2938 non-null   int64
12  Polio                                 2919 non-null   float64
13  Total expenditure                     2712 non-null   float64
14  Diphtheria                            2919 non-null   float64
15  HIV/AIDS                              2938 non-null   float64
16  GDP                                    2490 non-null   float64
17  Population                             2286 non-null   float64
18  thinness 1-19 years                   2904 non-null   float64
19  thinness 5-9 years                    2904 non-null   float64
20  Income composition of resources        2771 non-null   float64
21  Schooling                             2775 non-null   float64
dtypes: float64(16), int64(4), object(2)
memory usage: 505.1+ KB
```

```
Missing Values:
Country          0
Year             0
Status           0
Life expectancy  10
Adult Mortality  10
infant deaths    0
Alcohol          194
percentage expenditure  0
Hepatitis B      553
Measles          0
BMI              34
under-five deaths  0
Polio            19
Total expenditure  226
Diphtheria       19
HIV/AIDS         0
GDP              448
Population       652
thinness 1-19 years  34
thinness 5-9 years  34
Income composition of resources  167
Schooling        163
dtype: int64
```

Selected Data Shape: (2928, 22)

Training Data Shape: (2342, 21)

Testing Data Shape: (586, 21)

Random Forest Regression Model Trained Successfully

```
KDD RESULTS
KDD MAE : 1.0305682593856675
KDD RMSE : 1.6650461723245689
KDD R2 Score : 0.9679491570863064
```

Using file: Life Expectancy Data.csv

First 5 Rows:

	Country	Year	Status	Life expectancy	Adult Mortality	\
0	Afghanistan	2015	Developing	65.0	263.0	
1	Afghanistan	2014	Developing	59.9	271.0	
2	Afghanistan	2013	Developing	59.9	268.0	
3	Afghanistan	2012	Developing	59.5	272.0	
4	Afghanistan	2011	Developing	59.2	275.0	

	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	...	\
0	62	0.01	71.279624	65.0	1154	...	
1	64	0.01	73.523582	62.0	492	...	
2	66	0.01	73.219243	64.0	430	...	
3	69	0.01	78.184215	67.0	2787	...	
4	71	0.01	7.097109	68.0	3013	...	

	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population	\
0	6.0	8.16	65.0	0.1	584.259210	33736494.0	
1	58.0	8.18	62.0	0.1	612.696514	327582.0	
2	62.0	8.13	64.0	0.1	631.744976	31731688.0	
3	67.0	8.52	67.0	0.1	669.959000	3696958.0	
4	68.0	7.87	68.0	0.1	63.537231	2978599.0	

	thinness 1-19 years	thinness 5-9 years	Income composition of resources	\
0	17.2	17.3	0.479	
1	17.5	17.5	0.476	
2	17.7	17.7	0.470	
3	17.9	18.0	0.463	
4	18.2	18.2	0.454	

	Schooling
0	10.1
1	10.0
2	9.9
3	9.8
4	9.5

[5 rows x 22 columns]

Result: